Memory Management

Chapter 7

Basic Idea of Memory Management

- Subdividing memory to accommodate multiple processes
- Memory needs to be allocated to ensure a reasonable supply(合理供应) of ready processes to consume available processor time

Memory Management Requirements

- Relocation(重定位)
- Protection(保护)
- Sharing(共享)
- Logical Organization
- Physical Organization

Relocation

- Programmer does not know where the program will be placed in memory when it is executed
- While the program is executing, it may be swapped to disk and returned to main memory at a different location (relocated)
- Memory references must be translated in the code to actual physical memory address

Addressing Requirements for a Process

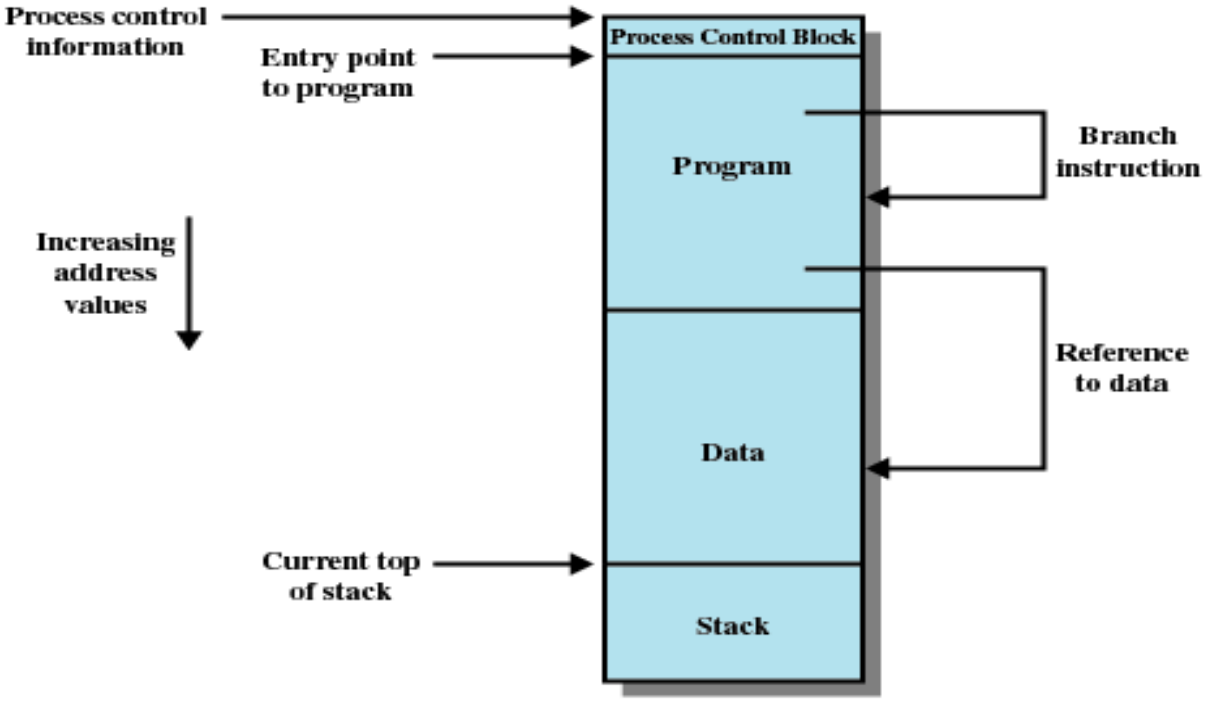


Figure 7.1 Addressing Requirements for a Process

Protection

- Processes should not be able to reference memory locations in another process without permission
- Impossible to check absolute addresses at compile time
- Address reference must be checked at run time
- Memory protection requirement must be satisfied by the processor (hardware) rather than the operating system (software)
  - Operating system cannot anticipate all of the memory references a program will make

Sharing

- Allow several processes to access the same portion of memory
  - Better to allow each process access to the same copy of the program rather than have their own separate copy

Logical Organization

- Programs are written in modules
- Modules can be written and compiled independently
- Different degrees of protection given to modules (read-only, execute-only)
- Share modules among processes

Physical Organization

- Memory available for a program plus its data may be insufficient
  - Overlaying allows various modules to be assigned the same region of memory
- Programmer does not know how much space will be available

Fixed Partitioning

- Equal-size partitions
  - Any process whose size is less than or equal to the partition size can be loaded into an available partition
  - If all partitions are full, the operating system can swap a process out of a partition
  - A program may not fit in a partition. The programmer must design the program with overlays(覆盖)
- Main memory use is inefficient. Any program, no matter how small, occupies an entire partition. This is called internal fragmentation(内零头/内部碎片).

Example of Fixed Partitioning

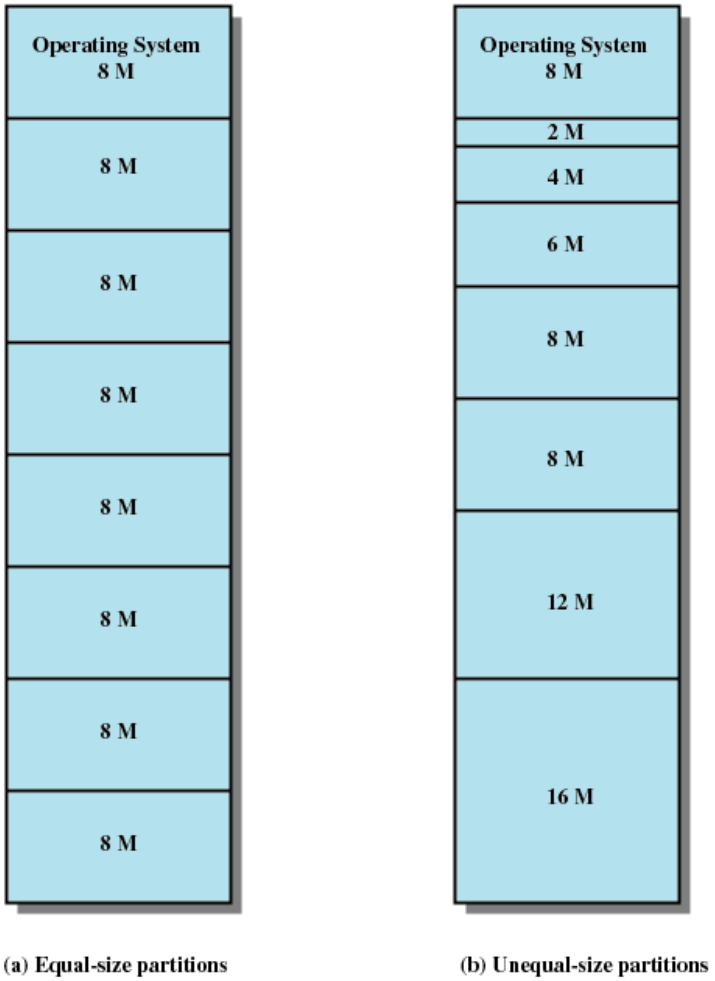


Figure 7.2 Example of Fixed Partitioning of a 64-Mbyte Memory

Placement Algorithm with Partitions

- Equal-size partitions
  - Because all partitions are of equal size, it does not matter which partition is used
- Unequal-size partitions
  - Can assign each process to the smallest partition within which it will fit
  - Queue for each partition
  - Processes are assigned in such a way as to minimize wasted memory within a partition

Memory Assignment for Fixed Partitioning

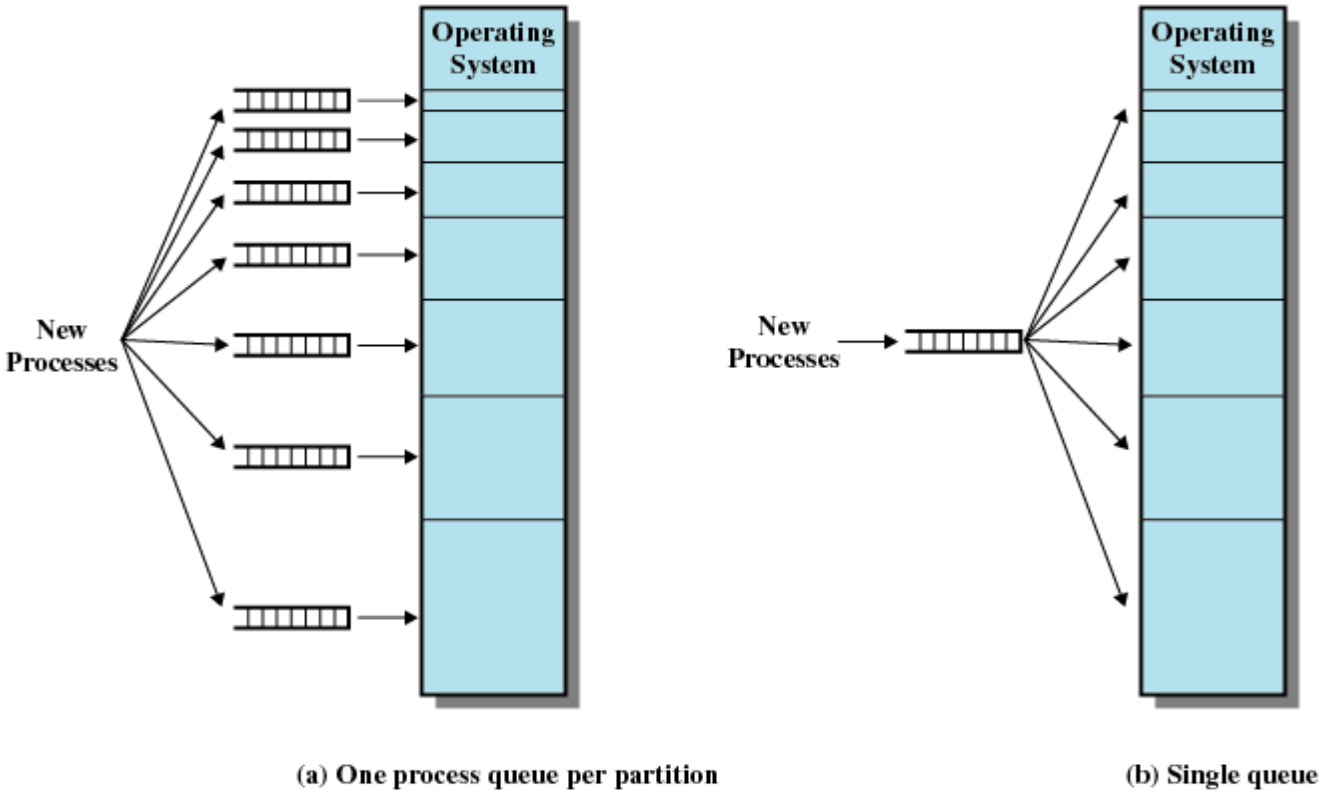


Figure 7.3 Memory Assignment for Fixed Partitioning

Dynamic Partitioning

- Partitions are of variable length and number
- Process is allocated exactly as much memory as required
- Eventually get holes in the memory. This is called external fragmentation (外零头/外部碎片)
- Must use compaction(紧缩) to shift processes so they are contiguous and all free memory is in one block

The Effect of Dynamic Partitioning

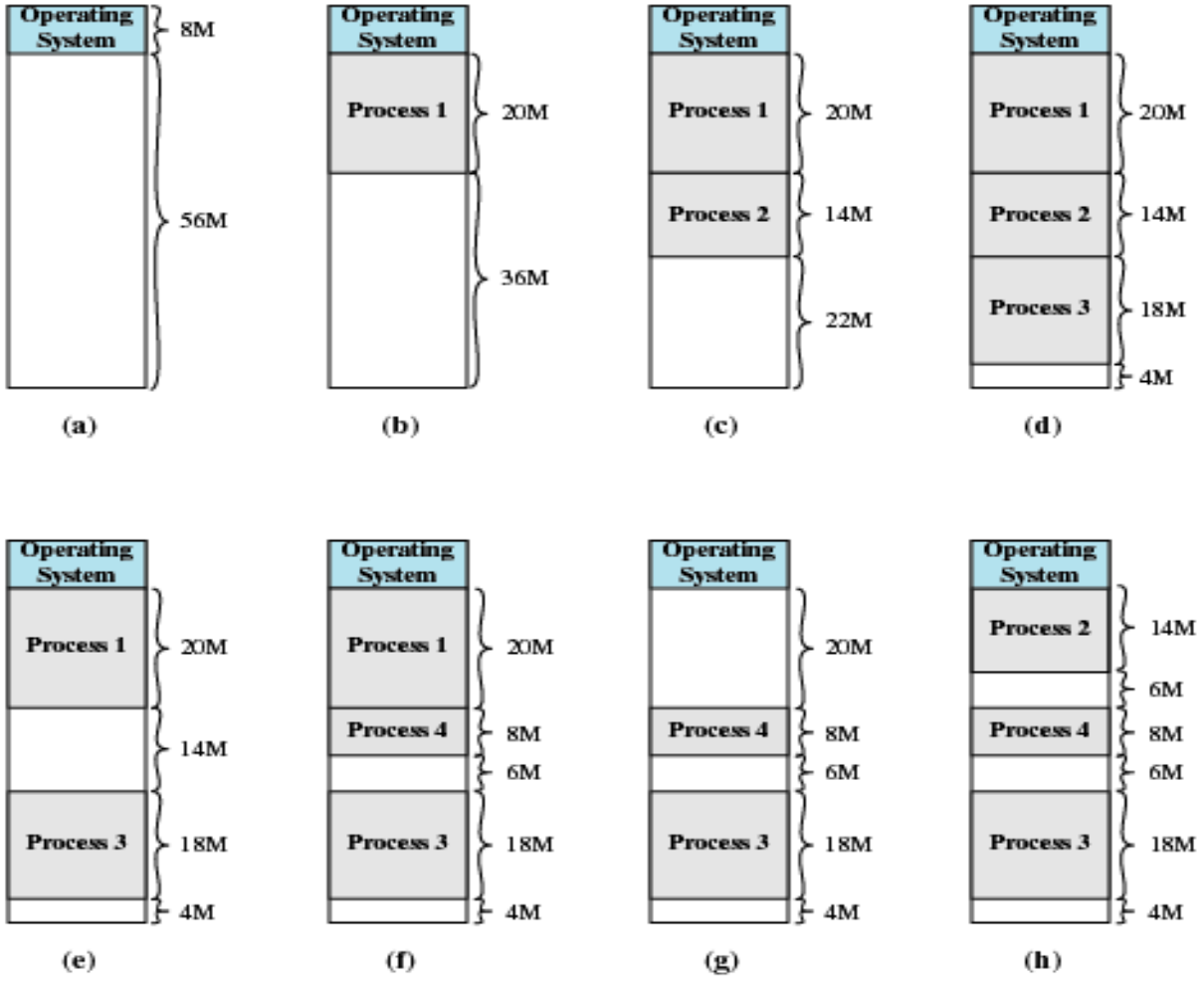


Figure 7.4 The Effect of Dynamic Partitioning

Dynamic Partitioning Placement Algorithm

- Best-fit(最佳适配) algorithm
  - Chooses the block that is closest in size to the request
  - Since smallest block is found for process, the smallest amount of fragmentation is left
  - Memory compaction must be done more often

Dynamic Partitioning Placement Algorithm

- First-fit(首次适配) algorithm
  - Scans memory from the beginning and chooses the first available block that is large enough
  - May have many process loaded in the front end of memory that must be searched over when trying to find a free block

Dynamic Partitioning Placement Algorithm

- Next-fit(下次适配)
  - Scans memory from the location of the last placement
  - More often allocate a block of memory at the end of memory where the largest block is found
  - The largest block of memory is broken up into smaller blocks
  - Compaction is required to obtain a large block at the end of memory

Example of Placement Algorithms

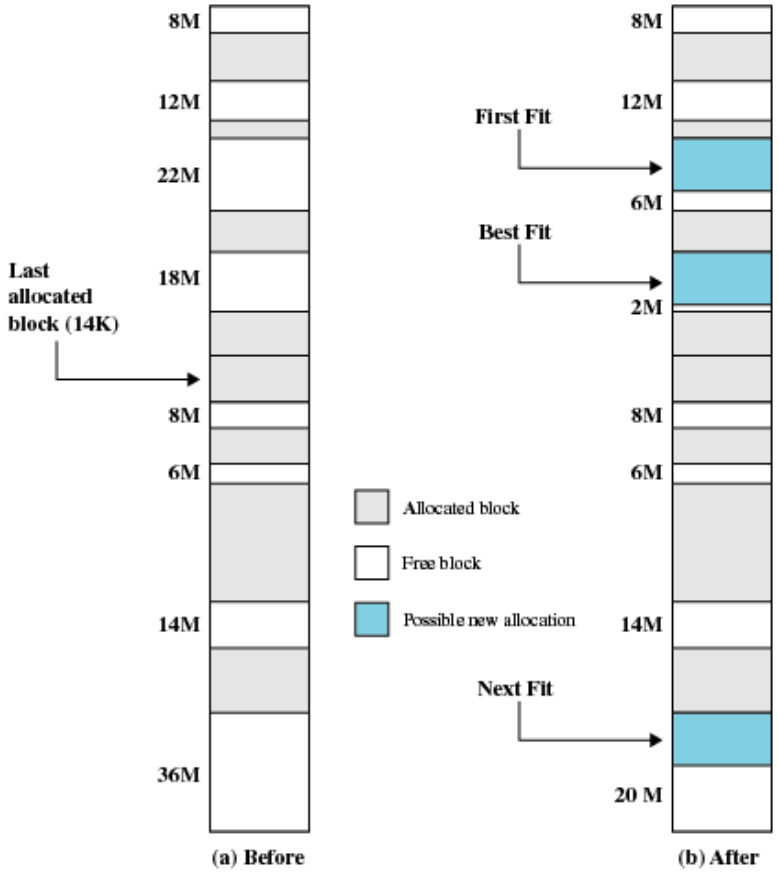


Figure 7.5 Example Memory Configuration Before and After Allocation of 16 Mbyte Block

Buddy System

- Entire space available is treated as a single block of 2^U
- If a request of size s such that $2^{U-1} < s <= 2^U$, entire block is allocated
  - Otherwise block is split into two equal buddies
  - Process continues until smallest block greater than or equal to s is generated

Example of Buddy System

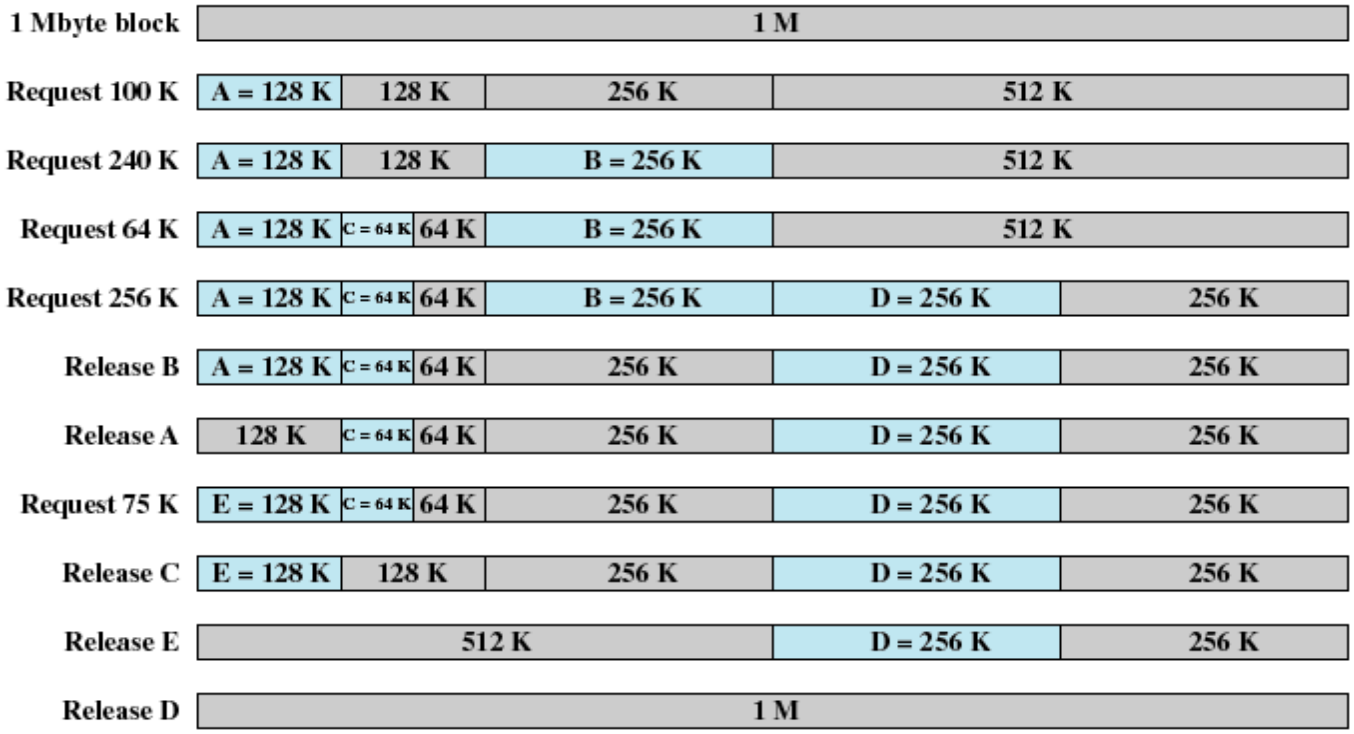


Figure 7.6 Example of Buddy System

Tree Representation of Buddy System

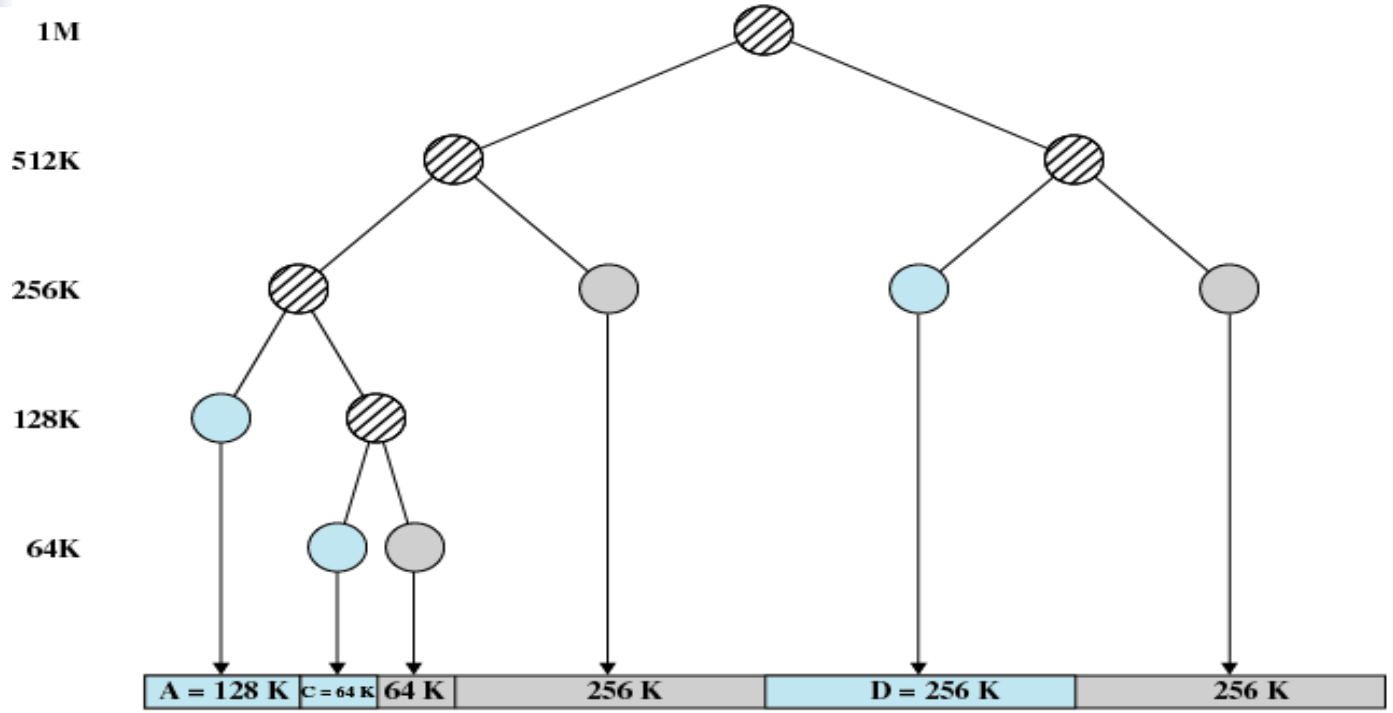


Figure 7.7 Tree Representation of Buddy System

Addresses

- Logical address
  - Reference to a memory location independent of the current assignment to memory
  - Translation must be made to the physical address
- Relative address
  - Address expressed as a location relative to some known point (BaseAddr : Offset)
- Physical address
  - The absolute address or actual location in main memory

Hardware Support for Relocation

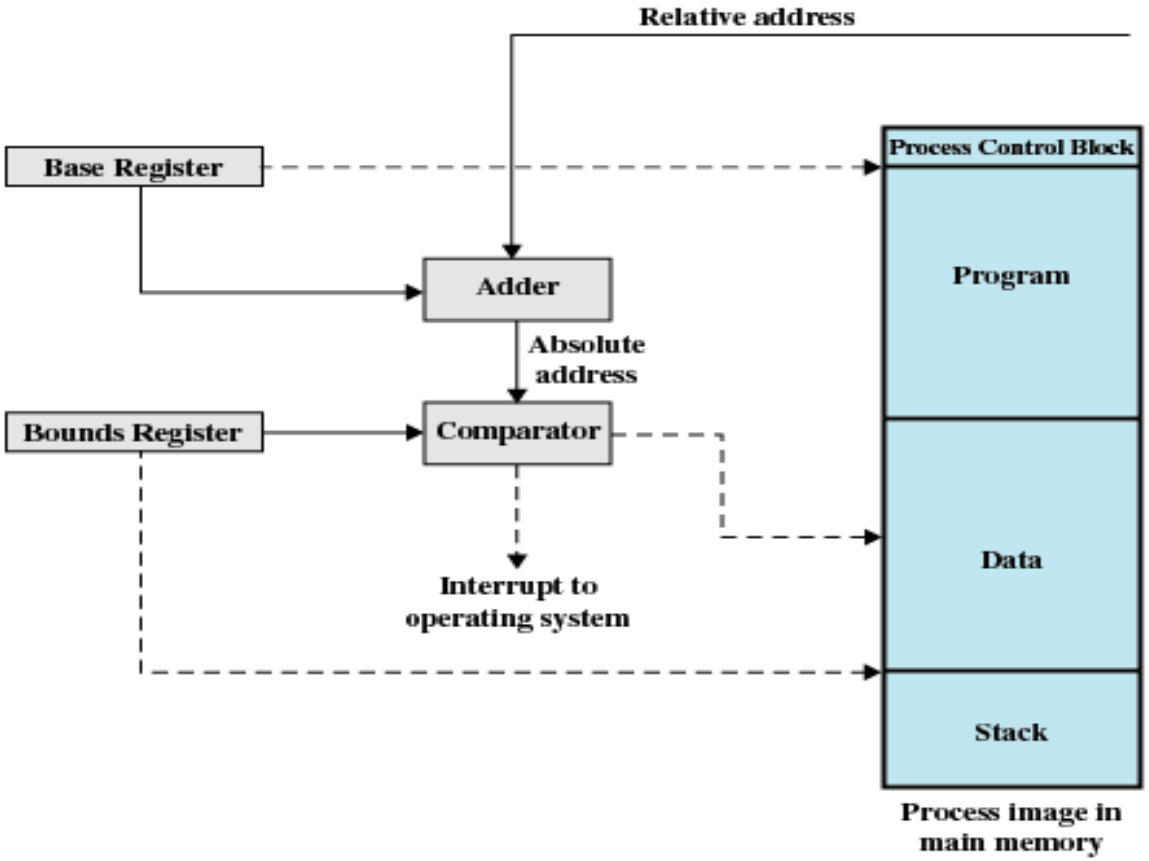


Figure 7.8 Hardware Support for Relocation

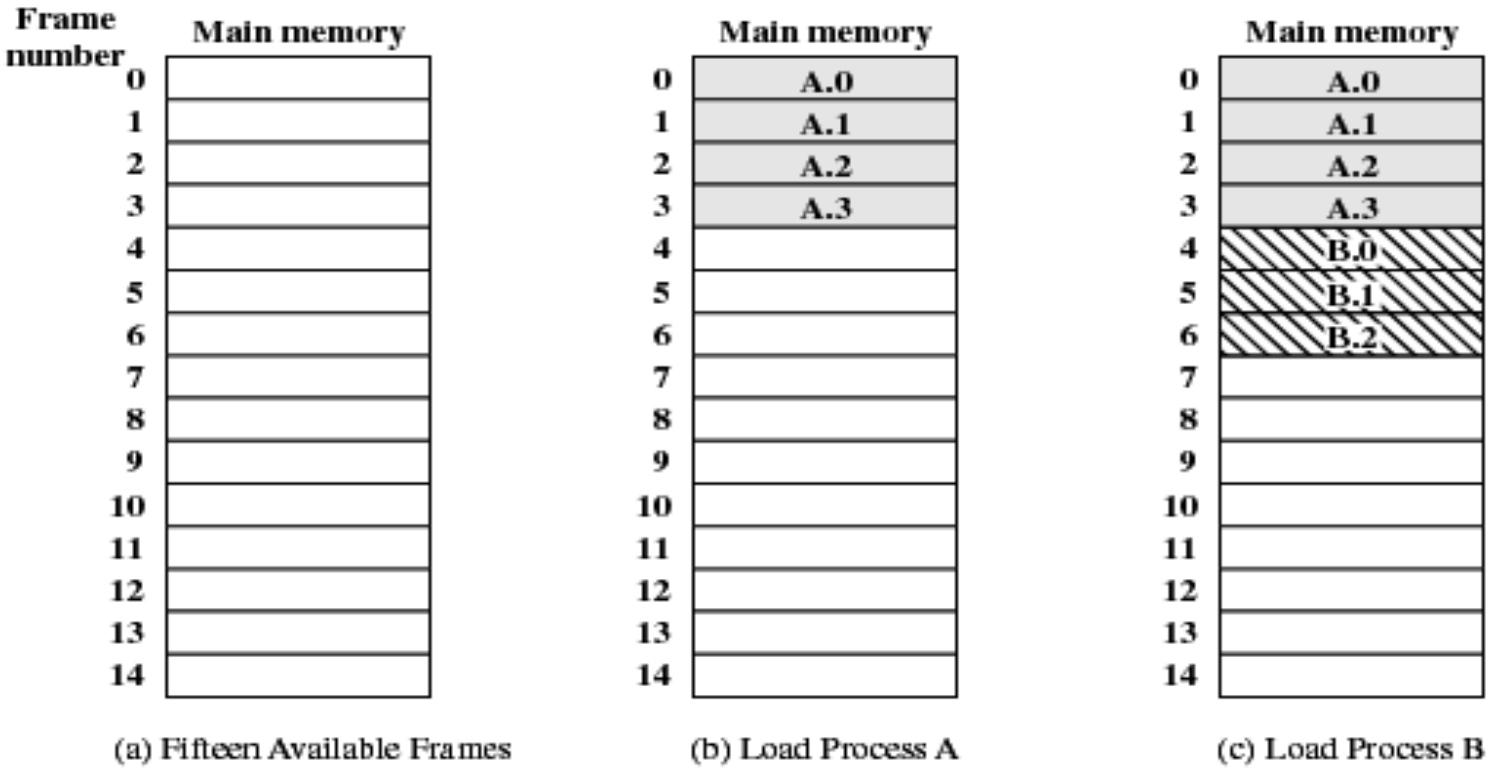
Registers Used during Execution

- Base register
  - Starting address for the process
- Bounds register
  - Ending location of the process
- These values are set when the process is loaded or when the process is swapped in
- The value of the base register is added to a relative address to produce an absolute address
- The resulting address is compared with the value in the bounds register
- If the address is not within bounds, an exception is generated to the operating system

Paging(分页)

- Partition memory into small equal fixed-size chunks and divide each process into the same size chunks
- The chunks of a process are called pages(页) and chunks of memory are called frames(页框)
- Operating system maintains a page table for each process
  - Contains the frame location for each page in the process
  - Memory address consist of a page number and offset within the page

Assignment of Process Pages to Free Frames



Assignment of Process Pages to Free Frames

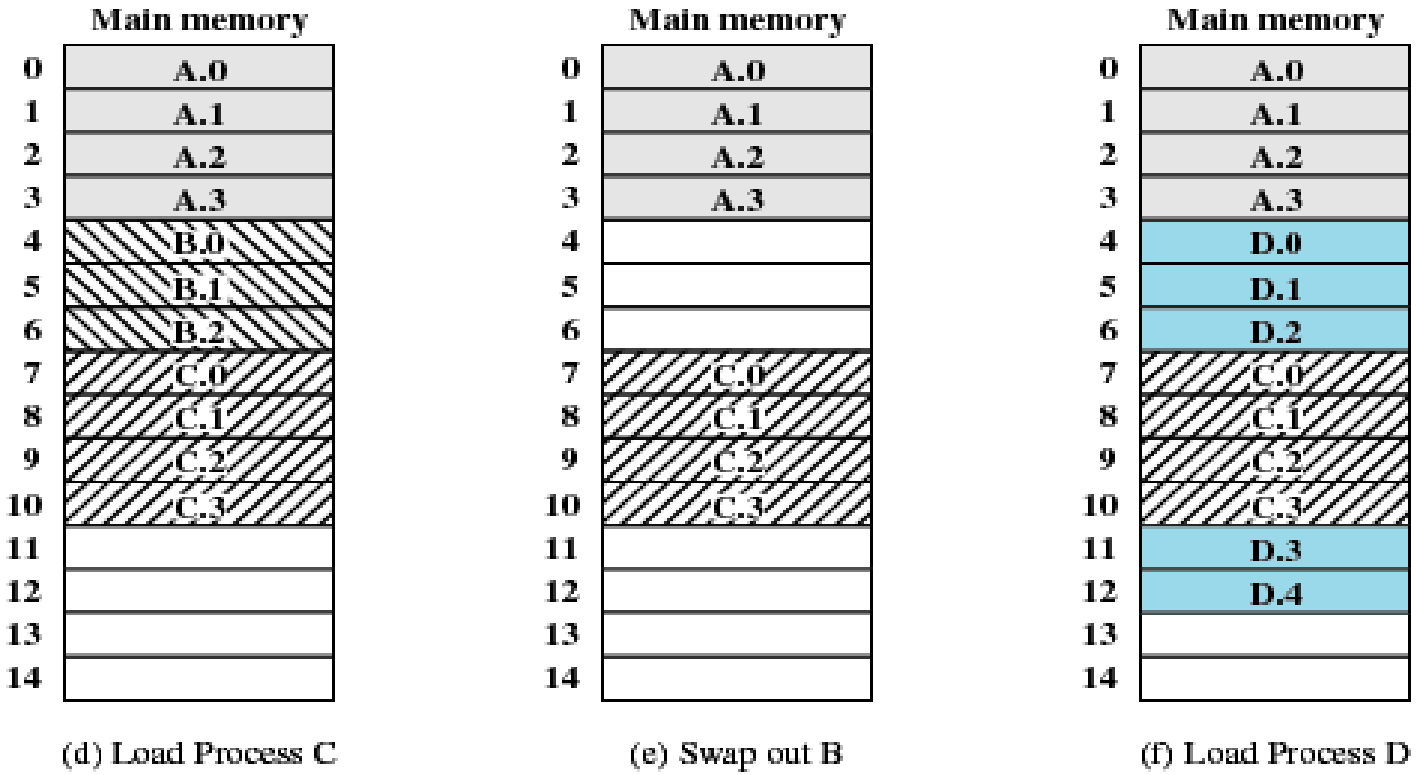


Figure 7.9 Assignment of Process Pages to Free Frames

Page Tables for Example

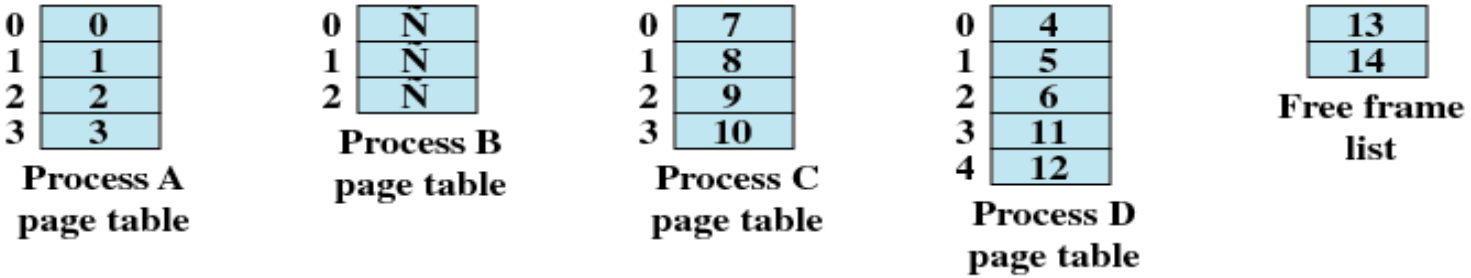


Figure 7.10 Data Structures for the Example of Figure 7.9 at Time Epoch (f)

Segmentation(分段)

- All segments of all programs do not have to be of the same length
- There is a maximum segment length
- Addressing consist of two parts - a segment number and an offset
- Since segments are not equal, segmentation is similar to dynamic partitioning

Logical Addresses

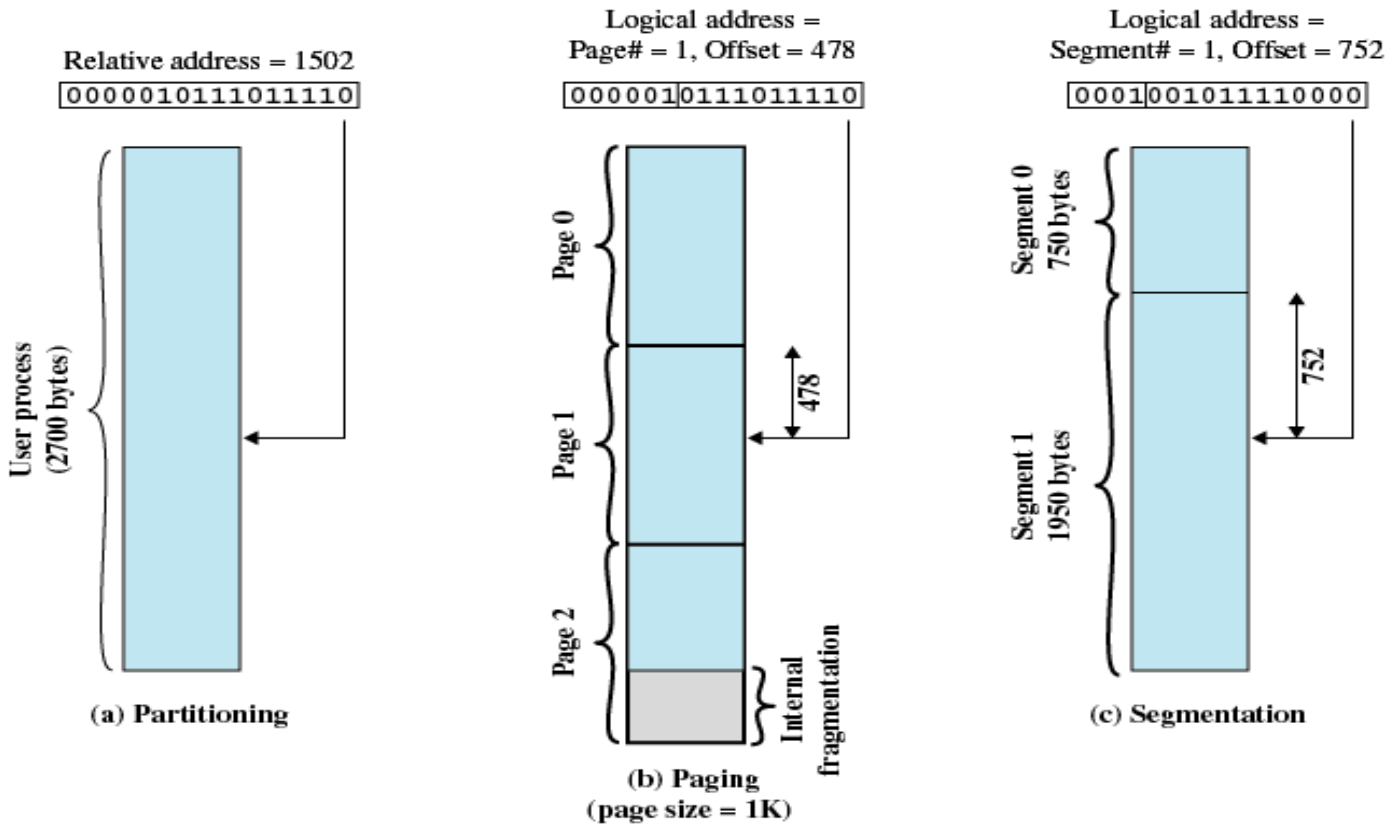
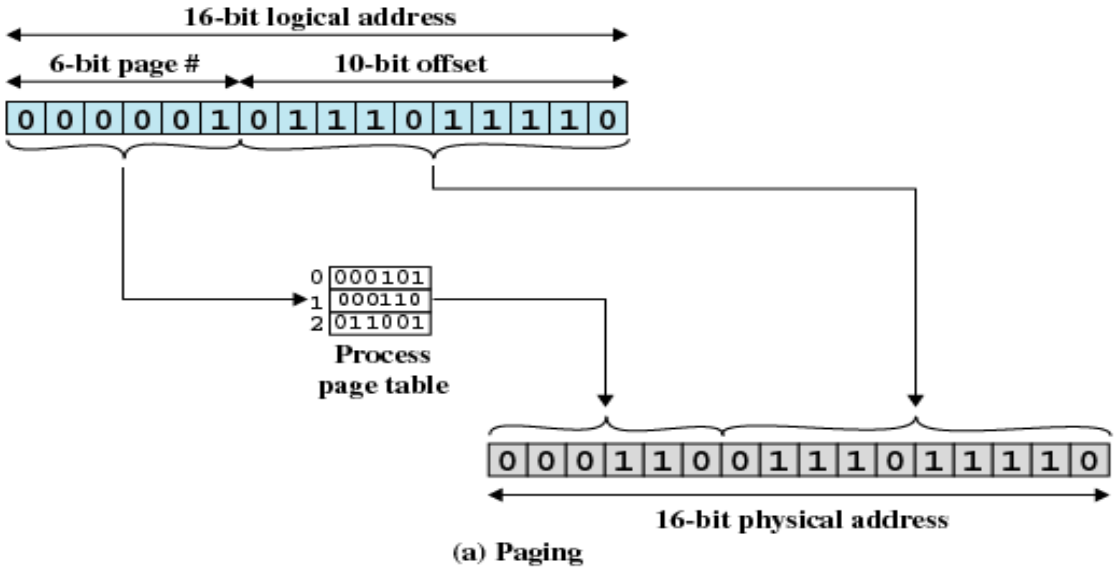


Figure 7.11 Logical Addresses

Paging



(a) Paging

Address Translation

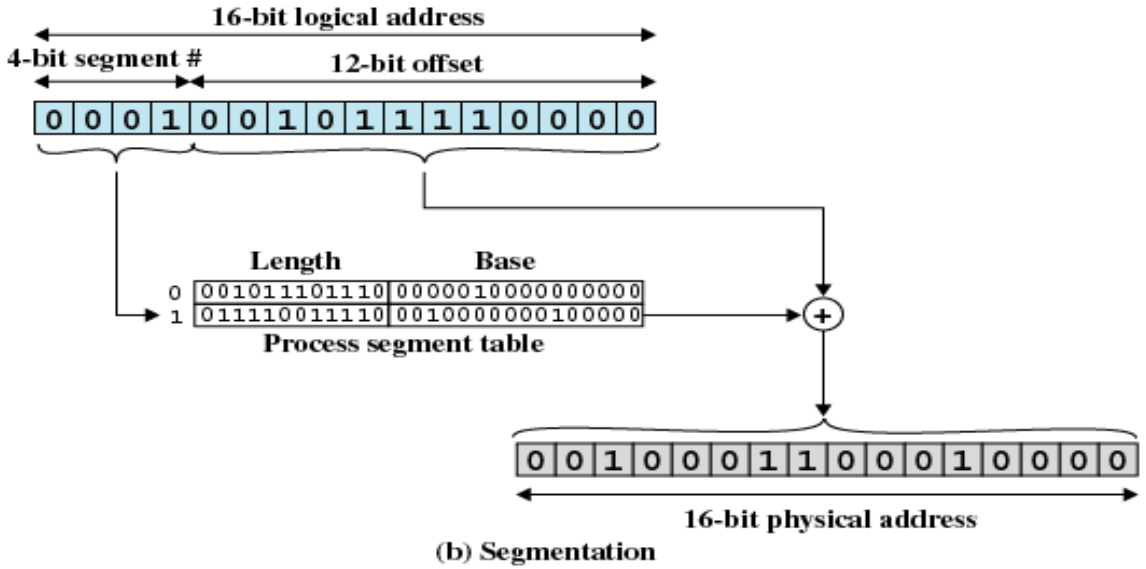


Figure 7.12 Examples of Logical-to-Physical Address Translation